

Consumption of coffee associated with reduced risk of liver cancer: a meta-analysis

Background Epidemiologic studies have reported inconsistent results regarding coffee consumption and the risk of liver cancer. We performed a meta-analysis of published case-control and cohort studies to investigate the association between coffee consumption and liver cancer. **Methods** We searched Medline, EMBASE, ISI Web of Science and the Cochrane library for studies published up to May 2012. We performed a meta-analysis of nine case-control studies and seven cohort studies. **Results** The summary odds ratio (OR) for high vs no/almost never drinkers was 0.50 (95% confidence interval (CI): 0.42–0.59), with no significant heterogeneity across studies ($Q = 16.71$; $P = 0.337$; $I^2 = 10.2\%$). The ORs were 0.50 (95% CI: 0.40–0.63) for case-control studies and 0.48 (95% CI: 0.38–0.62) for cohort studies. The OR was 0.38 (95% CI: 0.25–0.56) in males and 0.60 (95% CI: 0.33–1.10) in females. The OR was 0.45 (95% CI: 0.36–0.56) in Asian studies and 0.57 (95% CI: 0.44–0.75) in European studies. The OR was 0.39 (95% CI: 0.28–0.54) with no adjustment for a history of liver disease and 0.54 (95% CI: 0.46–0.66) after adjustment for a history of liver disease. **Conclusions** The results of this meta-analysis suggested an inverse association between coffee consumption and liver cancer. Because of the small number of studies, further prospective studies are needed.